GVM-Net: A GNN-Based Vessel Matching Network for 2D/3D Non-Rigid Coronary Artery Registration

Yankai Chen, Guanyu Li, Chunming Li, *Member, IEEE*, Wei Yu, Zehao Fan, Jingfeng Bai, *Member, IEEE*, and Shengxian Tu

***Abstract*— The registration of coronary artery structures from preoperative coronary computed tomography angiography to intraoperative coronary angiography is of great interest to improve guidance in percutaneous coronary interventions. However, non-rigid deformation and discrepancies in both dimensions and topology between the two imaging modalities present a challenge in the 2D/3D coronary artery registration. In this study, we address this problem by formulating it as a centerline feature matching task and propose a GNN-based vessel matching network (GVM-Net) to establish dense correspondence between different image modalities in an end-to-end manner. GVM-Net considers centerline points as nodes in graphs and effectively models the complex topological relationships between them through attention mechanisms and message passing. Furthermore, by incorporating redundant rows and columns into the matching matrix, GVM-Net can effectively handle inconsistencies in vascular structures. We also introduce the query-based nodes grouping module, which clusters nodes in the feature space to further explore the topological relationships. GVM-Net achieves an average F1-score of 89.74% with a mean pixel distance of 0.48 pixels on the synthetic dataset with 276 data pairs and an average F1-score of 83.35% with a mean error of 1.52 mm in 55 manually labeled clinical cases, both exceeding existing feature matching methods.**

***Index Terms*— 2D/3D non-rigid registration, feature matching, graph attention network, vessel graph matching.**

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Yankai Chen, Guanyu Li, Chunming Li, Wei Yu, Zehao Fan, and Jingfeng Bai are with the Department of Biomedical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: turingk@sjtu.edu.cn; guanyu.li@sjtu.edu.cn; lichunming0016@sjtu.edu.cn; heronet@sjtu.edu.cn; fzh_0615@sjtu.edu.cn; jfbai@sjtu.edu.cn).

Shengxian Tu is with the Department of Biomedical Engineering and the Institute of Medical Robotics, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: sxtu@sjtu.edu.cn).

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I. Introduction

CARDIOVASCULAR diseases, including ischemic heart disease, stroke, heart failure, peripheral arterial disease, and a range of cardiac and vascular conditions, are the leading cause of death globally and substantially impact quality of life [1]. Percutaneous coronary intervention (PCI) is the predominant and established treatment for patients with coronary artery disease. However, coronary angiography (CAG), the standard imaging technique for PCI, has notable limitations, including the difficulties in visualizing distal vessels within chronic total occlusion (CTO) lesions [2], [3], the loss of spatial information in the vessels, and an inability to determine plaque composition [2], [4]. These challenges can critically influence the assessment and success rate of PCI procedures.

Recent studies suggest that combining preoperative coronary computed tomography angiography (CTA) with intraoperative CAG can improve the success rates of PCI, especially in patients with CTO. [3], [5].However, there are still some unresolved challenges in the registration of coronary artery structures from CTA to CAG: **1) disparities in grayscale values** between the two imaging modalities, **2) deviations in projection parameters** caused by patient repositioning and system inaccuracies, **3) significant non-rigid deformations** caused by respiratory and cardiac motion, **4) inconsistencies in vascular structures**.

To address these challenges, many researchers have delved into this area and made significant advances [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29]. Due to the sparse nature of vascular structures, most registration methods depend on centerline features. These methods can be broadly classified into point-based, curve-based, and graph-based [30], [31], [32]. Previous studies indicate that graph-based registration, which leverages the rich topological information of vascular trees, yields superior results. However, existing methods in this field focus on the construction and matching of graph structures, as illustrated in Fig. 1. These methods still heavily rely on manually designed features and employ traditional iterative matching approaches, resulting in a limited number of matching points, slow registration speed, and difficulties in addressing inconsistent structures.

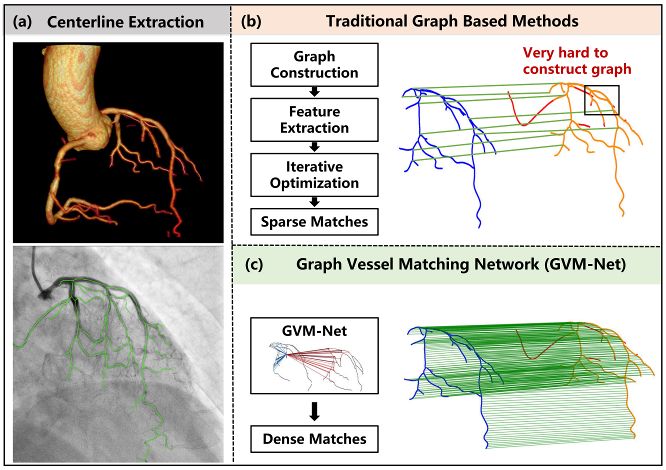

Fig. 1. Comparison of traditional graph-based methods and the **GNN-based Vessel Matching Network (GVM-Net)**. (a) Vessel centerline extraction from both CTA and CAG. (b) Traditional graph-based methods involve graph construction, feature extraction, and iterative optimization, resulting in complex and time-consuming sparse matches. (c) In contrast, GVM-Net skips graph construction and obtains the direct and dense matches with higher efficiency and quality.

Consequently, there is a pressing need for more effective vascular registration methods between CTA and CAG.

In this study, we address this problem by formulating it as a centerline feature matching task and propose a GNN-based vessel matching network (GVM-Net) to establish dense correspondence between different image modalities in an end-to-end manner. From the dense correspondence obtained from GVM-Net, we can derive closed-form solutions for both rigid and non-rigid registration, which improves accuracy and interpretability while avoiding iterative methods [33]. GVM-Net models centerline points as nodes within graph structures, effectively capturing complex topologies through attention mechanisms and message passing. By incorporating redundant rows and columns into the matching matrix, GVM-Net can effectively handle inconsistencies in vascular structures across modalities. The proposed query-based nodes grouping (QNG) module further refines this process by clustering nodes in feature space to better investigate their topological relationships. Furthermore, we propose an anatomy prior-based data synthesis method to generate realistic deformable vascular trees and dense correspondences for training and validation of the network.

The main contributions can be summarized as follows:

1) We propose formulating 2D/3D coronary artery registration as a centerline feature matching task and introduce GVM-Net, an end-to-end graph attention network, to establish dense correspondences across different imaging modalities. To the best of our knowledge, this study is the first to apply a graph attention network to the 2D/3D coronary artery registration problem.
2) We introduce query-based nodes grouping module (QNG) to augment the capabilities of GVM-Net in the context of centerline feature matching. This module efficiently clusters and updates node features within the feature space, leveraging topological information to improve matching accuracy.
3) We suggest a synthetic data generation method that incorporates anatomical knowledge to efficiently generate training and testing datasets. It addresses the challenge of obtaining data pairs and corresponding annotations in the field of non-rigid 2D/3D coronary artery registration.
4) Over a series of experiments and ablation studies, we showed that GVM-Net outperforms previous feature matching methods in 2D/3D non-rigid coronary artery registration.

II. Related Work

A. 2D/3D Coronary Artery Registration

The current coronary artery registration method can be broadly classified into intensity-based and feature-based [30], [31], [32]. Recently, deep learning techniques have also been introduced to address the challenges in coronary artery registration. In the following sections, we will introduce these methods sequentially.

1) Intensity-Based Methods: Intensity-based methods [6], [7] assess similarity by comparing grayscale differences between the digitally reconstructed radiograph (DRR) generated from the CTA volume and the CAG image. Feature-based methods simplify the representation of coronary arteries in centerline structures, focusing on preserving important details of the vascular tree while disregarding complex image backgrounds. This simplification improves the efficiency of registration and makes it the preferred choice [34].

2) Feature-Based Methods: Feature-based registration methods can be classified into point-based, curve-based, and graph-based [30]. **Point-based methods** represent vessel centerlines as discrete point sets. Registration is achieved by minimizing the distance between two sets of points, with common methods including iterative closest point (ICP) [35], coherent point drift (CPD) [36], and robust point matching (RPM) [37]. TPS-based methods such as TPS-RPM [22], TPS-L2 [27] and TPS-KC [28] utilize the thin-plate spline (TPS) model to handle non-rigid deformations by first optimizing point correspondences and then applying TPS to compute smooth transformations based on these correspondences. Advanced methods like PR-GLS [24] and RPM-L2E [25] further enhance robustness by leveraging shape context descriptors [29] to better handle noise and outliers during point matching. Serradell et al. [8] combined distance and orientation information, treating the matching problem as an optimal assignment problem and solving it using the Hungarian algorithm. Groher et al. [9] established soft correspondences between points using the Gold and Rangarajan algorithm [38]. Building upon Gaussian Mixture Models (GMM) [39], Baka et al. [10] constructed a four-dimensional GMM with coordinates and orientations, achieving registration by optimizing the L^2 distance error between two GMMs. **Curve-based methods** divide vessel centerlines into multiple segments at bifurcation points. The registration is achieved by comparing the shapes of the curves, considering the topological order between the points. Benseghir et al. [11], [12] extended the concept of ICP to the iterative closed curve (ICC), transforming point-to-point distances into distances between vessel segments and deepened their research by incorporating topological constraints within the ICC framework. In **graph-based methods**, the registration targets are points or curves on the centerline, but the registration process incorporates adjacency information from the graph as constraints. These methods transform the registration problem into a graph matching problem [13], [14], [15], [21]. Liu et al. [16] modeled the centerline structures of CTA and CAG as Directed Acyclic Graphs (DAGs) based on graph matching methods. Zhu et al. [17], [18] proposed the redundant graph matching algorithm (RGM), which accommodates the presence of pseudo-bifurcation points and completes the matching by constructing a matching matrix of nodes and edges. Yoon et al. [40] used the complete topological structure of CTA to restore the damaged topological structure of CAG, measuring similarity using the local bijection-pair method. Zhu et al. [21] proposed combining the Monte Carlo Tree Search method with manifold regularization to achieve accurate 3D/2D non-rigid vascular registration.

3) Learning-Based Method: In recent years, deep learning has significantly impacted coronary artery registration. Zhang et al. [19] introduced an unsupervised method based on point clouds, using the DGCNN network [41] to predict the deformation fields of the point clouds. Zhang et al. [26] introduced SPR-Net, which combines geometric and semantic features to obtain corresponding structural points. The method leverages 3D Thin Plate Splines (TPS) to compute non-rigid deformations, demonstrating strong performance in 3D coronary artery registration. Wu et al. [20] developed CAR-Net for topological segment-by-segment registration, using a 3D angular displacement field to improve registration precision.

B. Detector-Based Local Feature Matching

Local feature matching is essential in computer vision, encompassing feature detection, description, matching, and mismatch removal [42]. Successful correspondences allow for straightforward image registration using transformation models such as rigid, affine, or projective. Recent advancements in deep learning-based local feature matching, such as the introduction of SuperGlue by Sarlin et al. [43], have significantly progressed the field. SuperGlue uses transformers [44] and optimal transport [45] to address partial assignment problems, learning robust priors about scene geometry and camera motion that perform well across varied data domains. LightGlue, proposed by Philipp et al. [46], further enhances this approach by improving speed and effectiveness.

However, these methods, developed for natural image scenes, often fail to address the unique challenges of medical imaging, such as limited texture, low contrast, and non-rigid deformations. Furthermore, in the field of medical imaging, models like SuperGlue and LightGlue often struggle to obtain sufficient matches.

III. Method

We propose formulating 2D/3D coronary artery registration as a centerline feature matching task. Identifying a sufficient number of matching points facilitates both rigid and non-rigid registration, significantly enhancing speed, efficacy, and interpretability. The Methods section is organized into four main parts: First, the pre-processing steps for extracting vessel centerlines and preparing CTA and CAG images for feature matching. Second, the architecture of GVM-Net for establishing dense correspondences. Third, data synthesis and ground truth generation, which is crucial for creating synthetic datasets and necessary ground truth correspondences for computing the loss function. Finally, the loss function tailored to handle non-rigid deformations, ensuring robust vascular registration.

A. Pre-Processing

1) 2D/3D CenterLine Extraction: For vessel segmentation in CAG and CTA images, we employ the ResUnet++ [47] and 3D Unet [48] architectures, respectively. Unet is the most widespread image segmentation architecture due to its flexibility, optimized modular design, and success in all medical image modalities [49]. In our research, ResUnet++ is used for 2D segmentation in CAG, leveraging residual connections and Unet's skip connections to retain detailed vessel features. For 3D CTA data, the 3D U-Net captures both local detail and global vessel structure through its encoder-decoder architecture. Following segmentation, centerline images L_{cag}^{2d} for CAG and L_{cta}^{3d} for CTA were derived. For 2D centerline extraction in CAG, we apply the Zhang-Suen thinning algorithm [50], which produces thin, one-pixel-wide representations suitable for accurate 2D vessel tracking. In the case of 3D CTA, we employ the 3D Medial Axis Thinning algorithm [51], which reduces vessels to their centerlines while preserving the topology and spatial integrity of 3D structures. Additionally, to maintain essential morphological details, the grayscale values of the centerlines are mapped to their respective vessel radii. This approach retains information on vessel diameter, enhancing the subsequent matching process.

2) C-Arm Projection Model: We start by projecting 3D vessel centerlines L_{cta}^{3d} from CTA images onto a 2D plane to align with 2D centerlines L_{cag}^{2d} from CAG images. This process involves simulating the C-arm projection setup, where X-rays travel from a point source through cardiac structures to the detector. Referring to the approach outlined in reference [20], the light source's position $\boldsymbol{p}$ can be mathematically expressed as follows:

$$\boldsymbol{p} = \left(\frac{d_{xc} \tan \alpha}{p_{\text{base}}}, -\frac{d_{xc}}{p_{\text{base}}}, \frac{d_{xc} \tan \beta}{p_{\text{base}}} \right), \tag{1}$$

where $p_{\text{base}} = \sqrt{1 + \tan^2 \alpha + \tan^2 \beta}$. d_{xc} is the distance of source to patient, α and β is primary angle and secondary angle. The camera intrinsic matrix K can be obtained as:

$$K = \begin{bmatrix} d_{xd}/\text{ spacing} & 0 & u_0 \\ 0 & d_{xd}/\text{ spacing} & v_0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{2}$$

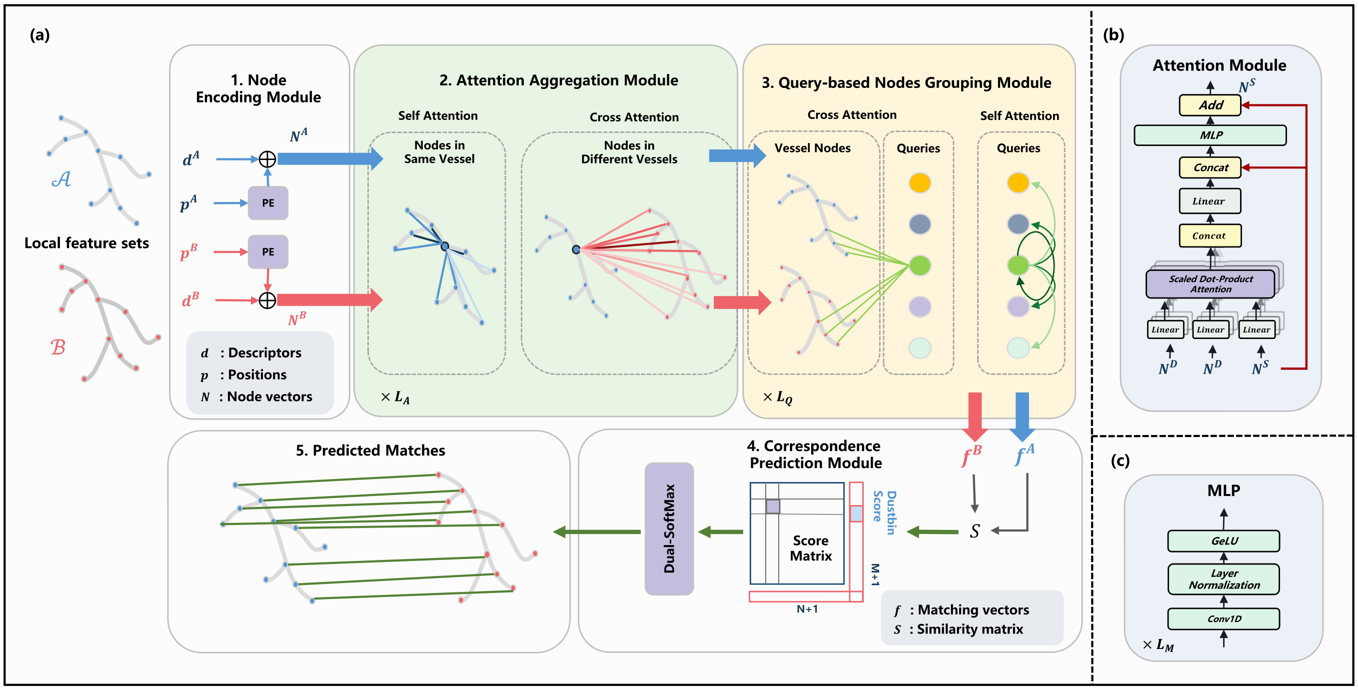

Fig. 2. Overview of GVM-Net and its components. (a) The architecture of GVM-Net. GVM-Net comprises a graph attention network backbone and a correspondence prediction module. The graph attention network backbone consists of node encoding, attention aggregation, and query-based nodes grouping modules. (b) The attention module used in self- and cross-attention. (c) MLP used in node encoding and attention aggregation module.

where d_{xd} is the distance of source to detector, u_0 and v_0 is the coordinates of principal point. Assuming the light source coordinate system has the light source position p as the center, pointing towards the center of the detector plane as the z-axis, and the axes parallel to the width and height of the plane are the x-axis and y-axis, respectively, then the coordinate axes in the light source coordinate system are represented as:

$$x_{\text{axis}} = \left[-\frac{py}{\sqrt{px^2+py^2}}, \frac{px}{\sqrt{px^2+py^2}}, 0\right],$$
$$z_{\text{axis}} = -\frac{p}{d_{xc}},$$
$$y_{\text{axis}} = \left[\frac{px \cdot pz}{y_{\text{base}}}, \frac{py \cdot pz}{y_{\text{base}}}, -\frac{px^2+py^2}{y_{\text{base}}}\right], \tag{3}$$

where $y_{\text{base}} = \sqrt{(px^2+py^2)^2 + px^2 \cdot pz^2 + py^2 \cdot pz^2}$. So we can obtain R and T:

$$R = [x_{\text{axis}} \quad y_{\text{axis}} \quad z_{\text{axis}}]^\top,$$
$$T = \begin{bmatrix} np \cdot \text{dot}(-p, x_{\text{axis}}) \\ np \cdot \text{dot}(-p, y_{\text{axis}}) \\ np \cdot \text{dot}(-p, z_{\text{axis}}) \end{bmatrix}. \tag{4}$$

Through the following process, project points from the CT to the CAG plane for dimension unification:

$$x = K[R \mid T]X. \tag{5}$$

Following this projection step, L_{cta}^{3d} is converted into L_{cta}^{2d} for analysis.

3) Dense Keypoint Extraction: The pre-trained SuperPoint [52] network is employed as a keypoint detector for L_{cta}^{2d} and L_{cag}^{2d}. SuperPoint is a fully-convolutional neural network designed for efficient, pixel-level interest point detection and description. It uses a self-supervised method to enhance geometric consistency. When applied to centerline images, where pixel values outside the vessel skeleton are zeros, the detected keypoints naturally concentrate along the centerlines. Additionally, by adjusting the non-maximum suppression (NMS) distance and the number of keypoints, we can control both the density and the total number of keypoints along the centerlines. The keypoints and descriptors for L_{cta}^{2d} are represented as P_{cta} and D_{cta}, respectively, and for L_{cag}^{2d} as P_{cag} and D_{cag}, providing a robust foundation for GVM-Net's feature matching process.

B. GVM-Net

1) Problem Definition: After pre-processing, we extract local features from L_{cta}^{2d} and L_{cag}^{2d}. These features, labeled as P_{cta} and P_{cag} for positional data, and D_{cta} and D_{cag} for visual descriptors, originate from the respective 2D centerlines. Each local feature includes a 2D point position $p_i = (x_i, y_i)$, normalized within the unit square $[0, 1]^2$, along with an associated visual descriptor $d_i \in \mathbb{R}^d$. The feature sets from L_{cta}^{2d} and L_{cag}^{2d} comprise m and n local features, respectively, indexed by sets $\mathcal{A} := \{1, \ldots, m\}$ and $\mathcal{B} := \{1, \ldots, n\}$. We introduce GVM-Net, a network designed to predict a subset of potential correspondences between these feature sets. Specifically, GVM-Net aims to produce a set of correspondences $M = \{(i, j)\} \subseteq \mathcal{A} \times \mathcal{B}$. This is achieved by computing a soft partial assignment matrix $P \in [0, 1]^{m \times n}$, which represents the probabilistic matching between the local features of L_{cta}^{2d} and L_{cag}^{2d}. The correspondences are determined by identifying significant probabilities in the matrix P, indicating the probability of matching the features between the two centerlines.

2) Node Encoding Module: To emphasize the importance of positional information, the node encoding module is introduced to encode the position vectors p_i and then fuse them with the descriptor vectors d_i to obtain the initial node vectors n_i. The node encoding module is implemented through a multilayer perceptron (MLP), as illustrated in (6).

$$\mathbf{n}_i = \mathbf{d}_i + \mathrm{MLP}_{\mathrm{enc}}(\mathbf{p}_i). \tag{6}$$

The encoded nodes are subsequently labeled as $N^A \in \mathbb{R}^{n\times d}$ and $N^B \in \mathbb{R}^{m\times d}$, where n and m are the number of nodes in each subset, and d is the feature dimension. The two subsets are used in following modules.

3) Attention Aggregation Module: Recognizing the positional and structural relationships among keypoints within and between modalities is essential for accurate matching. The attention aggregation module updates node features in two stages: self-attention and cross-attention. In the self-attention step, each node's features are refined by aggregating information from other nodes in the same set using residual message passing. The updated N^A is expressed as:

$$N^{A'} = N^A + \mathrm{Message}(N^A, N^A). \tag{7}$$

Next, a cross-attention step is performed to further enrich the node features by incorporating information from the other node set. The updated N^A after cross-attention is represented as:

$$N^{A''} = N^{A'} + \mathrm{Message}(N^{A'}, N^{B'}). \tag{8}$$

And the update process for N^B follows the same steps as N^A. In both updates, messages for each node are computed using an MLP applied to the concatenation of the node's original feature and the aggregated attention information. Specifically, for each node i, the message is calculated as:

$$\mathrm{Message}_i(N^A, N^B) = \mathrm{MLP}\left([\mathbf{n}_i^A \,\|\, \mathrm{Attention}_i(N^A, N^B)]\right), \tag{9}$$

where $[\cdot\,\|\,\cdot]$ denotes concatenation, and $\mathrm{Attention}_i(N^A, N^B)$ is computed using the multi-head attention mechanism. In the multi-head mechanism, multiple attention heads are calculated independently and then concatenated to obtain a richer representation, each attention head $\mathrm{Attention}_i^k$ is calculated as:

$$\mathrm{Attention}_i^k(N^A, N^B) = \sum_{j=1}^{m} \alpha_{ij} * \mathbf{v}_j^k. \tag{10}$$

Here, $\alpha_{ij} = \mathrm{softmax}_j\left(\mathbf{q}_i^{k\top}\mathbf{k}_j^k/\sqrt{d}\right)$, is the attention weight between nodes. And $\mathbf{q}_i^k$. $\mathbf{k}_j^k$, and $\mathbf{v}_j^k$ are the query, key and value vectors for the k-th head, obtained through linear projections of the node features from n_i^A and n_j^B, respectively.

This two-step process, first aggregating internal information through self-attention and then incorporating external information through cross-attention, ensures that each node feature is enriched comprehensively by both intra-set and inter-set relationships, facilitating robust feature representation. The multi-head attention mechanism enhances this process by allowing the model to capture different types of relationships from multiple perspectives, ultimately improving the quality of feature aggregation and propagation. The process will be repeated L_A times to obtain updated N^A and N^B vectors.

4) Query-Based Nodes Grouping Module: Node relationships can be also expressed and updated using a clustering-like approach. This aligns with human intuition, where the corresponding vascular segments are identified first, followed by matching points within these segments. So we propose the QNG module to simulate this process. The module also operates in two steps: cross-attention (Eq. 11) and self-attention (Eq. 12):

$$Q' = Q + \mathrm{Message}(Q, N)$$
$$N' = N + \mathrm{Message}(N, Q), \tag{11}$$
$$Q'' = Q' + \mathrm{Message}(Q', Q'), \tag{12}$$

where the set of queries $Q \in \mathbb{R}^{q\times d}$ is randomly initialized and q represents the number of queries. These queries serve as learnable cluster centers. The node set $N \in \mathbb{R}^{(n+m)\times d} = [N^A \,\|\, N^B]$ consists of the concatenation of two subsets: $N^A \in \mathbb{R}^{n\times d}$ and $N^B \in \mathbb{R}^{m\times d}$.

This two-step process first aggregates information from nodes and queries through cross-attention, followed by refining the queries with self-attention, enabling effective clustering representation. The process will be repeated L_Q times to obtain the final updated N^A, N^B and N^Q vectors.

5) Correspondence Prediction Module: After obtaining the updated node features, we first perform a linear transformation to obtain the final matching vectors.

$$\mathbf{f}_i = \mathbf{W} \cdot {}^{(l)}\mathbf{n}_i + \mathbf{b}. \tag{13}$$

Then the similarity matrix $S^p \in \mathbb{R}^{n\times m}$ can be obtained by the dot product of the matching vectors:

$$\mathbf{S}_{ij}^p = \left(\mathbf{f}_i^A\right)^\top \mathbf{f}_j^B. \tag{14}$$

To address the issue of unmatched points due to inconsistent topological structures, we add "dustbin" rows and columns to the similarity matrix S^p and fill them with learnable parameters. The "dustbin" allows unmatched points to be explicitly assigned, preventing errors caused by points that do not have corresponding matches in the other modality. Then, we apply dual-softmax, which has been applied in previous local feature matching works [46], [53] on all rows and column. The final matching matrix $\overline{P} \in \mathbb{R}^{(n+1)\times(m+1)}$ is calculated by taking the geometric mean of the row-wise and column-wise softmax values:

$$\overline{\mathbf{P}} = \sqrt{\mathrm{softmax}_{\mathrm{row}}(\mathbf{S}^p) \odot \mathrm{softmax}_{\mathrm{col}}(\mathbf{S}^p)}. \tag{15}$$

Given this final assignment matrix, we retain the mutual nearest neighbors with matching scores above the specified threshold.

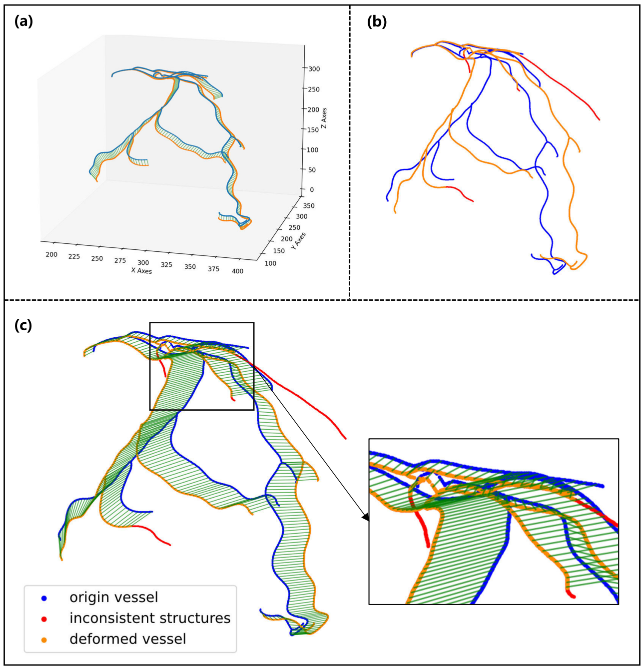

Fig. 3. Visualization of synthetic deformed coronary trees and their correspondence. (a) A pair of 3D deformed coronary trees. (b) 2D projections of the deformed trees. (c) A pair of trees along with their correspondences. In all figures, the displacement of the ostia point is eliminated to better display the generated non-rigid deformation.

6) Deep SuperVision: GVM-Net iteratively performs attention aggregation to update the features of the nodes and finally obtain the matching matrix. Due to the lightweight of dual-softmax, we can calculate and supervise the matching matrix at the intermediate layer of attention aggregation. Specifically, at the intermediate layers of the attention aggregation module where supervision is desired, the QNG module is first employed to refine the features, then the correspondence prediction module is used to predict the matching matrix for supervision.

C. Data Synthesis and Ground Truth Generation

Due to the presence of non-rigid deformations, even manual annotations by doctors may not accurately capture the matching relationships. To address this issue, we introduce a data generation method infused with anatomical knowledge to create synthetic datasets and ground truth correspondences, as illustrated in Fig. 3. This method plays a key role in generating data pairs for both training and validation.

We start by using 3D coronary artery centerlines extracted from CT scans to synthesize data pairs. The coronary tree is treated as a set of points $\mathbf{C} \in \mathbb{R}^{\mathbf{n}\times\mathbf{3}}$. Given the absence of overlaps, the topological structure of C is well-defined. Each path of the centerline, from the ostium to the leaf point, are identified. We assume that during vascular deformation, the distal points along the same centerline are influenced by their preceding proximal points. To formalize this, we define an ancestor matrix $\mathbf{AM} \in \mathbb{R}^{\mathbf{n}\times\mathbf{n}}$:

$$AM_{ij} = \begin{cases} 1 & \text{if } C_j \text{ is a preceding node of } C_i, \\ 0 & \text{otherwise.} \end{cases} \tag{16}$$

Using AM, the preceding nodes for each point in C are determined. Each point C_i is then expressed in local spherical coordinates $CS_i = (r_i, \theta_i, \phi_i)$, where:

$$\begin{aligned} r_i &= \|C_i - C_{\text{pre}(i)}\|, \\ \theta_i &= \arctan\left(\frac{y_i - y_{\text{pre}(i)}}{x_i - x_{\text{pre}(i)}}\right), \\ \phi_i &= \arccos\left(\frac{z_i - z_{\text{pre}(i)}}{r_i}\right). \end{aligned} \tag{17}$$

To introduce non-rigid deformations, we generate local displacements for each point. We assume that the distance between adjacent points remains nearly constant during deformation. Therefore, we set the radial displacement $\Delta r_i = 0$, preserving the relative positions of neighboring points along the centerline. The angular offsets $\Delta\theta_i$ and $\Delta\phi_i$ are computed as follows:

$$\begin{aligned} \Delta\theta_i &= d \cdot U(a, b) \cdot \sin(U(0, \frac{\pi}{2})), \\ \Delta\phi_i &= d \cdot U(a, b) \cdot \sin(U(0, \frac{\pi}{2})), \end{aligned} \tag{18}$$

where $U(a, b)$ is a uniform distribution, and d controls the deformation scale. The local displacement in Cartesian coordinates $\Delta Ci = (\Delta x_i, \Delta y_i, \Delta z_i)$ is calculated as:

$$\begin{aligned} \Delta x_i &= (r_i + \Delta r_i) \cdot \sin(\theta_i + \Delta\theta_i) \cdot \cos(\phi_i + \Delta\phi_i), \\ \Delta y_i &= (r_i + \Delta r_i) \cdot \sin(\theta_i + \Delta\theta_i) \cdot \sin(\phi_i + \Delta\phi_i), \\ \Delta z_i &= (r_i + \Delta r_i) \cdot \cos(\theta_i + \Delta\theta_i). \end{aligned} \tag{19}$$

The total displacement for each point is obtained by aggregating the local displacements of its ancestor nodes using the ancestor matrix AM:

$$\mathbf{C}' = \mathbf{C} + \mathbf{AM} \cdot \Delta\mathbf{C}. \tag{20}$$

Then we simulate the projection process of a C-arm machine to mimic projection parameter errors caused by patient movements and equipment variability between CTA and CAG. The resulting 2D centerline points C_{2d} are obtained as:

$$\mathbf{C}'_{\mathbf{2d}} = Proj(\mathbf{C}', \mathbf{P} + \Delta\mathbf{P} \cdot \mathbf{s}), \tag{21}$$

where P represents the projection parameters, ΔP is the perturbation, and s controls the scale of projection perturbation. Finally, we simulate topological inconsistencies by adding noise and removing vessel segments. This method ensures that our training and testing datasets accurately represent the complexities and challenges encountered in real-world vascular images, laying a solid foundation for the training and evaluation of our algorithms.

D. Loss Function

To address the challenges of feature matching in non-rigid and low-contrast scenarios commonly found in medical images, we introduce two loss components: L_{Focal} and L_{Logdist}.

The L_{Focal} emphasizes challenging matches by reducing the importance of high-confidence matches (where $\bar{P}_{i,j}$ is close to 1) and focusing on points with lower matching confidence.

The parameter γ controls the degree of down-weighting for high-confidence matches. In this paper, we set γ to 2.

$$L_{\text{Focal}} = -\left(1-\overline{\mathbf{P}}_{i,j}\right)^{\gamma}\sum_{(i,j)\in\mathcal{M}}\log\overline{\mathbf{P}}_{i,j} - \left(1-\overline{\mathbf{P}}_{i,N+1}\right)^{\gamma}\sum_{i\in\mathcal{I}}\log\overline{\mathbf{P}}_{i,N+1} - \left(1-\overline{\mathbf{P}}_{M+1,j}\right)^{\gamma}\sum_{j\in\mathcal{J}}\log\overline{\mathbf{P}}_{M+1,j}. \quad (22)$$

The L_{Logdist}, on the other hand, introduces pixel distance information to guide the matching process more effectively. Given that A_i's ground truth matching point is B_g, the distance $D_{i,j}$ represents the Euclidean distance between B_g and B_j. This can be expressed as:

$$D_{i,j} = \|B_m - B_j\|. \quad (23)$$

Then the L_{Logdist} can be formulated as:

$$L_{\text{Logdist}} = \frac{1}{|\mathcal{M}|}\sum_{(i,j)\in\mathcal{M}}\overline{\mathbf{P}}_{i,j}\log(\mathbf{D}_{i,j}+1). \quad (24)$$

By supervising the matching process based on pixel distances, this term provides smoother guidance compared to only supervising the matching matrix $\bar{P}$. The log function enhances sensitivity to medium-to-low distances, promoting precise alignment for closer matches, which is especially important in handling non-rigid deformations. By combining these two losses, the overall objective function L_{Final} is:

$$L_{\text{Final}} = \alpha L_{\text{Focal}} + \beta L_{\text{Logdist}}, \quad (25)$$

where α and β are hyperparameters. In this paper, we set α to 1, and the setting of β is compared in the ablation experiment section.

IV. Experiments and Results

A. DataSet

Our dataset is divided into two parts. The first part, referred to as DataSyn, consists of data pairs generated by our data synthesis method. Each data pair includes two deformed centerlines along with their correspondences, which were used to train and evaluate matching algorithms. DataSyn comprises 840 three-dimensional vascular trees, which were randomly divided into 696 cases for training, 98 for validation, and 46 for testing. During the training phase, we adopted the online data generation strategy.

To ensure fair test results, we augmented the test set by applying different levels of deformation and projection errors. For easy cases, we set the deformation level to $d = 0$ and the projection error level to $s = 0.5$, representing minimal deformation with moderate projection error. For medium cases, we set $d = 0.5$ and $s = 0.5$, reflecting moderate deformation and projection error. For hard cases, we set $d = 1$ and $s = 1$, representing significant deformation and higher projection error. These adjustments increased the test set to a total of 276 cases.

For external validation, independent clinical datasets with coronary CTA and CAG prior to any coronary intervention were provided by OLV Clinic, Aalst, Belgium and by Fujian Medical University Union Hospital, Fuzhou, China. The ethics committees of the two hospitals approved the retrospective analysis of the datasets, with patients providing written informed consent. For each patient, the vascular centerlines of the preoperative CTA and CAG images were extracted using the mentioned methods. We annotated the matching points for the left and right coronary arteries in sequences with optimal projection angles. After quality control, 55 cases were included for validation, with an average of 35 annotated matches per case. The detailed information of the datasets are presented in Table I. To ensure sufficient sample sizes for meaningful analysis, we combined data from centers 2, 3, 4, and 5, where case numbers were limited, into a single group. As a result, the 55 cases from the five centers were divided into two groups for subsequent analysis, with Group 1 containing 30 cases and Group 2 containing 25 cases.

TABLE I
Information of Clinical Dataset Used for Testing

		Case Number	Modality	Manufacturer	Spatial Resolution (mm)
Group 1	Center 1	30	CT	SIEMENS	0.39 * 0.39 * 0.60
			XA	GE	0.20 * 0.20
Group 2	Center 2	15	CT	SIEMENS	0.37 * 0.37 * 0.75
			XA	GE	0.20 * 0.20
	Center 3	4	CT	SIEMENS	0.30 * 0.30 * 0.60
			XA	Philips	0.17 * 0.17
	Center 4	2	CT	SIEMENS	0.33 * 0.33 * 0.75
			XA	Philips	0.34 * 0.34
	Center 5	4	CT	TOSHIBA	0.35 * 0.35 * 0.50
			XA	TOSHIBA	0.29 * 0.29

B. Experimental Setup

1) Evaluation Metrics: To quantitatively evaluate matching algorithms, we calculated metrics that include precision, recall, F1-score and mean pixel distance (mPD) between prediction and ground truth. When calculating accuracy, for the synthetic dataset, a predicted matching point falling within a 3-pixel range of the actual matching point is considered correct; meanwhile, for the clinical dataset, this threshold is set at 1.5 mm based on empirical considerations.

$$mPD = \frac{1}{N}\sum_{i=1}^{N}\sqrt{(x_i^{pred}-x_i^{gt})^2+(y_i^{pred}-y_i^{gt})^2}. \quad (26)$$

2) Implementation Details: We conducted quantitative and qualitative evaluations on DataSyn within a PyTorch framework, using PyTorch 1.10.1, CUDA 11.8.89, Python 3.7.16, Ubuntu 18.04 LTS and a NVIDIA 2080Ti GPU. The input data included feature points and descriptors sourced from SuperPoint, applying NMS with a threshold of 2 and limiting to 1024 feature points, supplemented with random noise where necessary. For GVM-Net, the intermediate feature representations were 256-dimensional. We set the attention aggregation module to $L_A = 11$ layers and the QNG module to $L_Q = 3$ layers. The deep supervision was applied at the 9th and 11th layers of the attention aggregation module. We implemented the multi-head attention with 4 heads, and

TABLE II

COMPARISON WITH SOTA METHODS. FINETUNED INDICATES TRAINING DEEP MATCHER WITH DATASYN

	Methods/Metrics	mPD ↓	Precision ↑	Recall ↑	F1-Score ↑
Conventional Matchers	BF	5.00	65.33	60.65	62.54
	FLANN	5.60	69.63	49.84	57.56
	GMS	3.30	61.66	62.90	61.07
Deep Matchers	SuperGlue^{indoor}	3.04	58.51	74.86	64.00
	SuperGlue^{outdoor}	1.95	71.60	82.17	76.15
	LightGlue	2.38	68.87	81.19	74.02
Finetuned Deep Matchers	SuperGlue^{finetuned}	1.20	79.14	88.73	83.40
	LightGlue^{finetuned}	0.80	84.36	90.67	87.23
Proposed Method	**GVM-Net**	**0.48**	**87.36**	**92.47**	**89.74**

TABLE III

COMPARISON WITH SOTA METHODS AMONG DIFFERENT SAMPLES. THE VALUES IN THE TABLE REPRESENT THE mPD (PIXEL) METRIC

	Methods/Metrics	Left	Right	Easy	Medium	Hard
Conventional Matchers	BF	6.37	3.60	3.08	3.39	8.46
	FLANN	6.68	4.48	3.56	3.66	9.47
	GMS	3.63	2.96	2.34	2.57	4.96
Deep Matchers	SuperGlue^{indoor}	3.21	2.88	2.61	2.66	3.92
	SuperGlue^{outdoor}	2.24	1.70	1.46	1.53	2.93
	LightGlue	2.72	2.06	1.69	1.77	3.73
Finetuned Deep Matchers	SuperGlue^{finetuned}	1.34	1.07	0.84	0.88	1.91
	LightGlue^{finetuned}	0.83	0.76	0.34	0.48	1.56
Proposed Method	**GVM-Net**	**0.50**	**0.46**	**0.33**	**0.39**	**0.72**

the final matching threshold was set to 0.1. The training phase used the AdamW optimizer, initiated at a learning rate of 1e-5. We used Cosine Annealing Warm Restarts with a decay factor of 2 over 10 epochs, gradually decreasing the learning rate to a minimum of 5e-8. The model was trained with a batch size of 8 over 500 epochs.

C. Comparisons With State-of-the-Art Methods

To evaluate the performance of GVM-Net in vessel matching tasks, we compared it with traditional matchers including brute force (BF), FLANN [54], and GMS [55], along with deep learning-based matchers including SuperGlue [43] and LightGlue [46]. We also fine-tuned SuperGlue and LightGlue by setting $d = 0$ (no deformation) and varying $s = 0.1, 0.5, 1.0$, to represent easy, medium and hard cases, respectively. These projection-based image pairs were derived from the original coronary dataset to ensure the models could learn from the vascular structure without introducing non-rigid deformations. This approach aligns with their original algorithm design, which generates image pairs only using affine transformations. We evaluated these approaches using various qualitative and quantitative metrics. This detailed evaluation allowed us to understand the effectiveness of GVM-Net in vessel matching tasks, providing valuable insights into its strengths and potential in this domain.

1) Quantitative Evaluation: Table II presents the performance metrics of various methods on the DataSyn dataset, while Fig. 4 provides a detailed illustration of their performance, highlighting the distribution of F1-score and mPD. It is evident that deep learning-based methods significantly outperform traditional methods in all metrics. These methods also demonstrate substantial improvements after fine-tuning on coronary datasets. GVM-Net outperforms traditional methods and other deep learning-based methods in all metrics, achieving an mPD of 0.48 and an F1-score of 89.74%. In addition, we present metrics for these methods in both left and right coronary arteries, as well as in different complexity cases, in Table III. The results confirm the consistent and outstanding performance of GVM-Net in challenging samples, including those involving significant deformations.

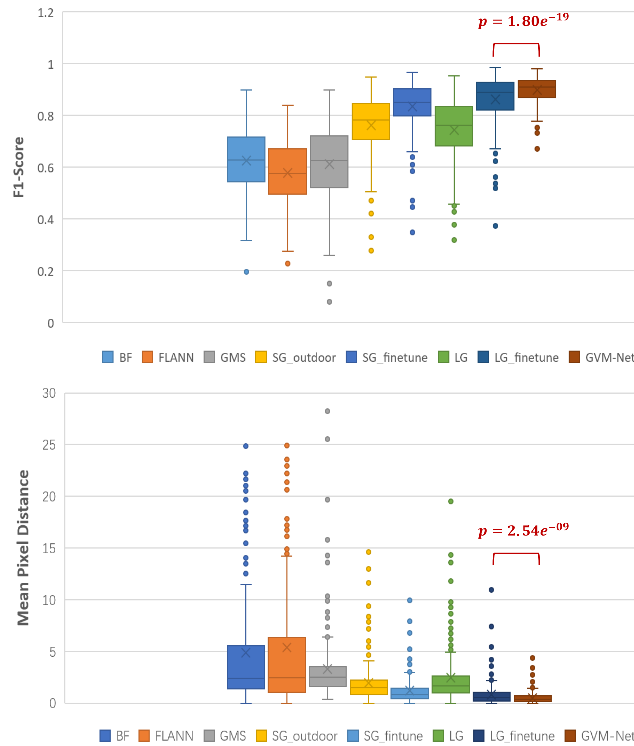

Fig. 4. The distribution of F1-score and mPD (pixel) on DataSyn. The paired t-test results indicate a significant difference between fine-tuned LightGlue and GVM-Net.

TABLE IV

ABLATION EXPERIMENT. DSV: DEEP SUPERVISION. DEFORM: TRAINING USING THE PROPOSED DATA GENERATION METHOD. LOSS: L_{FINAL} WITH $\alpha = 1.0, \beta = 0.3$. QNG: QUERY-BASED NODES GROUPING MODULE

Methods/Metrics	mPD ↓	Precision ↑	Recall ↑	F1-Score ↑
BaseLine	0.98	82.86	88.80	85.52
+ DSV	0.88(-0.10)	84.30(+1.44)	87.90(-0.90)	85.84(+0.32)
+ Radius	0.80(-0.08)	84.52(+0.22)	86.75(-1.15)	85.38(-0.46)
+ Deform	0.60(-0.20)	85.40(+0.98)	88.31(+1.56)	86.71(+1.33)
+ Loss	0.52(-0.08)	86.56(+1.16)	90.37(+2.06)	88.32(+1.61)
+ QNG	**0.48(-0.04)**	**87.36(+0.80)**	**92.47(+2.10)**	**89.74(+1.42)**

2) Qualitative Evaluation: We present several examples to compare the performance of different matching algorithms (Fig. 5). GMS and LightGlue are selected to represent conventional and deep matching techniques. For cases with significant non-rigid deformations, GVM-Net can deliver robust and reliable matches, while GMS and LightGlue produce sparse or incorrect results.

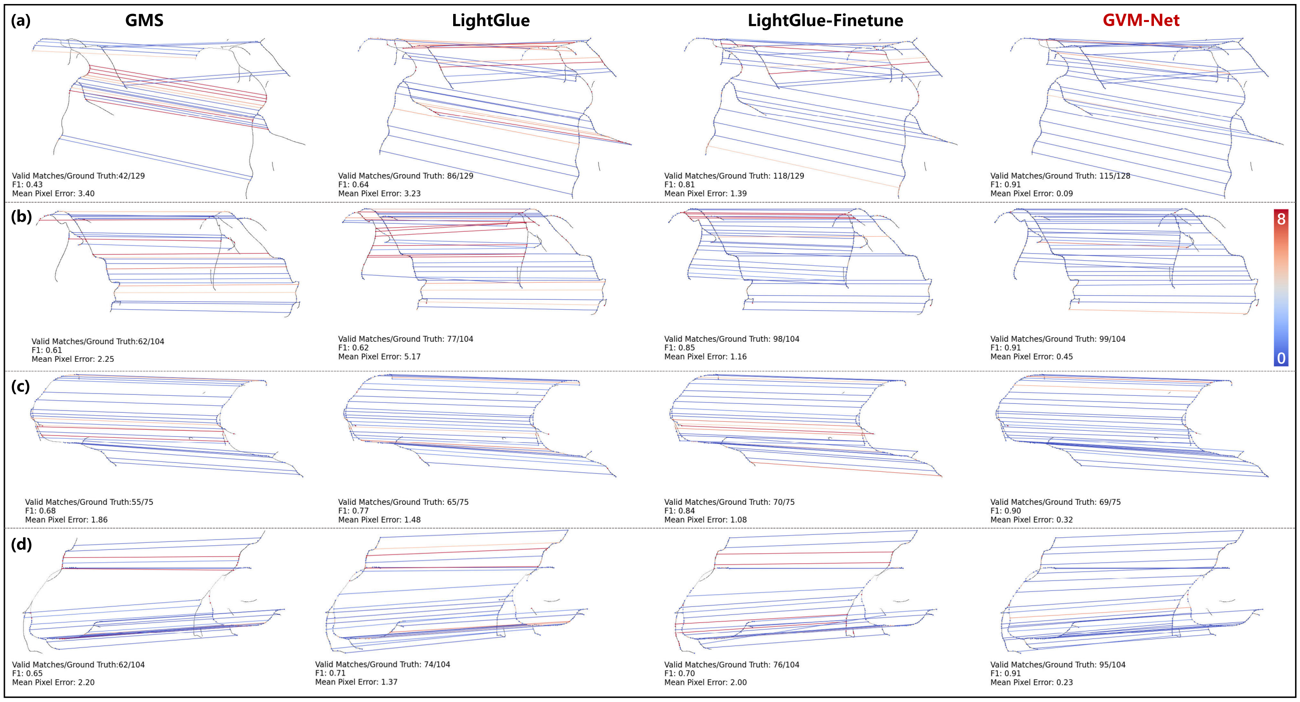

Fig. 5. Qualitative results on DataSyn. (a) and (b) show matching results of the left coronary artery, while (c) and (d) demonstrate matching results of the right coronary artery in two examples. The color bar indicates the mPD. Zoom in on the figure to view the details.

D. Comparison With Point Cloud Registration Methods

To demonstrate the robustness and efficiency of GVM-Net compared to traditional point-based registration methods, we conducted experiments using a synthesized dataset with varying deformation levels $d = 0.3, 0.5, 1.0$, and a small projection error level $s = 0.01$. GVM-Net was evaluated alongside 5 well-known point cloud registration algorithms: Mixture TPS-RPM [22], TPS-L2 [27], TPS-KC [28], RPM-L2E [25] and PR-GLS [24]. All thess methods used keypoints extracted by SuperPoint, with a maximum of 1024 points and NMS $= 1$. Each algorithm was employed to compute the deformation fields. For GVM-Net, we used the predicted matching points and applied the TPS deformation model to generate the deformation field. These deformation fields were then applied to the ground truth point sequences to calculate the registration error.

Quantitative metrics (Table. V) and qualitative results (Fig. 6) consistently demonstrate that GVM-Net outperforms these traditional point cloud methods in both accuracy and efficiency. GVM-Net shows greater tolerance to noise and outliers, while providing more precise point matching. These findings highlight GVM-Net's significant advantages over traditional point cloud methods, especially in complex scenarios.

TABLE V
COMPARISON BETWEEN GVM-NET AND TRADITIONAL POINT CLOUD REGISTRATION METHODS

Method	Time (s)	Recall Ratio			
		2 pixel	4 pixel	5 pixel	8 pixel
Mixture TPS-RPM	27.29 ± 16.66	20.23	44.78	53.90	71.83
PR-GLS	20.64 ± 27.82	27.62	51.50	59.68	75.61
RPM-L2E	13.10 ± 10.64	30.70	58.38	66.86	81.52
TPS-KC	21.63 ± 16.26	40.66	64.56	71.22	82.54
TPS-L2	23.29 ± 17.10	41.12	65.17	71.94	82.95
TPS + GVM-Net	**0.25 ± 0.13**	**75.64**	**88.80**	**91.20**	**95.02**

E. Robustness of GVM-Net to Incomplete Centerline Segmentations

GVM-Net performs matching based on feature points and descriptors from 2D centerline images, without requiring explicit vascular topology modeling. As a result, it remains robust even when vessel segmentations are imperfect, such as in cases of discontinuities or errors.

To evaluate GVM-Net's robustness in cases where vessel centerline extractions are imperfect, we conducted additional tests on the dataset with simulated segmentation errors. Specifically, we introduced random 8×8 pixel dropouts in each image, setting the pixel value of specific segments of the vessel centerlines to zero. GVM-Net was then tested on these modified image pairs, and the results were compared with those from the original dataset. Despite the induced discontinuities, GVM-Net's performance remained robust. Both qualitative visualizations (Fig. 7) and quantitative metrics (Table VI) shows no significant difference, confirming that GVM-Net can handle incomplete segmentations without a significant impact on its performance, which is highly desirable in practical applications.

F. Ablation Study

To assess GVM-Net's effectiveness, we conducted ablation studies detailed in Table IV. These studies demonstrate

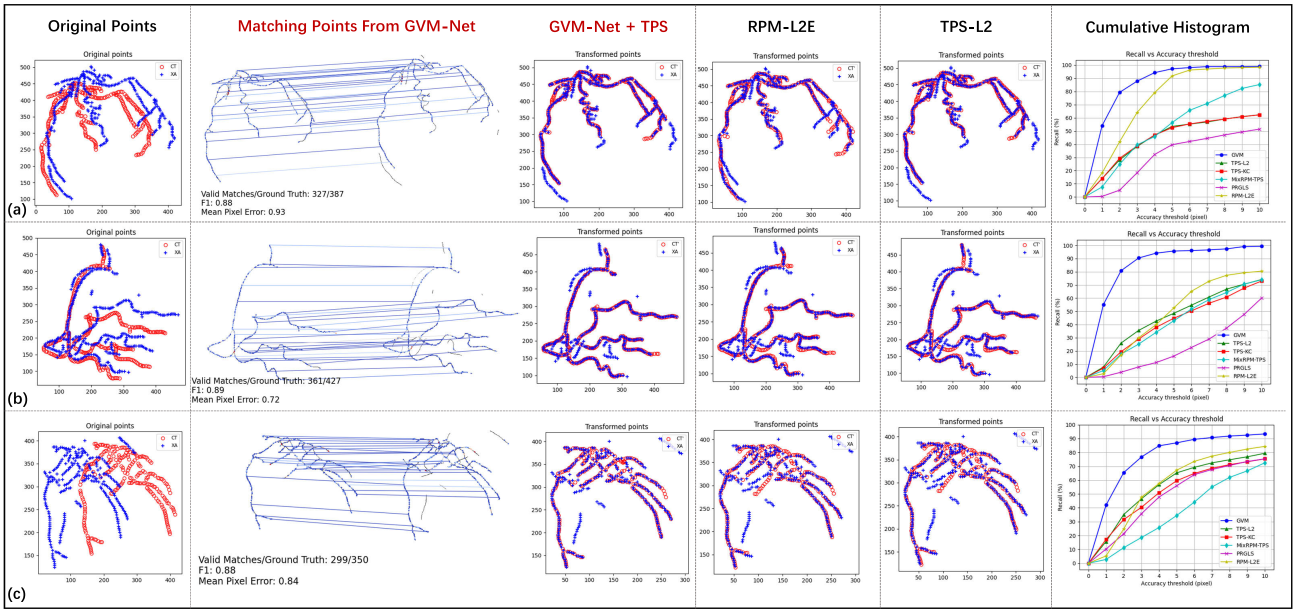

Fig. 6. Comparison between GVM-Net and traditional point cloud registration methods. We select the two top-performing traditional methods, RPM-L2E and TPS-L2, for visualization. Additionally, the cumulative histograms are provided to present the quantitative results across all methods. For GVM-Net, we show both the matching results and the TPS-based transformation. The results demonstrate that GVM-Net achieves more accurate and robust matching, with fewer errors in the transformed points. GVM-Net particularly outperforms traditional methods in complex scenarios with large deformations and outliers.

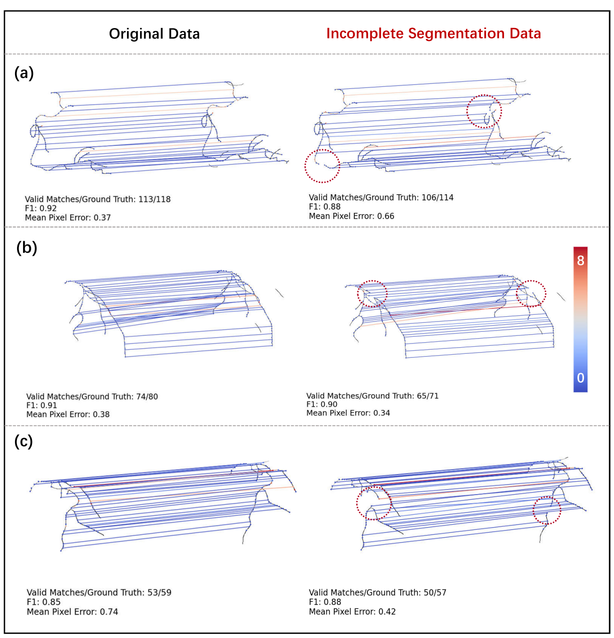

Fig. 7. Visual comparison of GVM-Net's performance on original and incomplete segmentation dataset. The red circles highlight regions where vessel centerlines are broken in the simulated incomplete segmentation data. Despite these imperfect centerline segmentations, GVM-Net's performance remains robust.

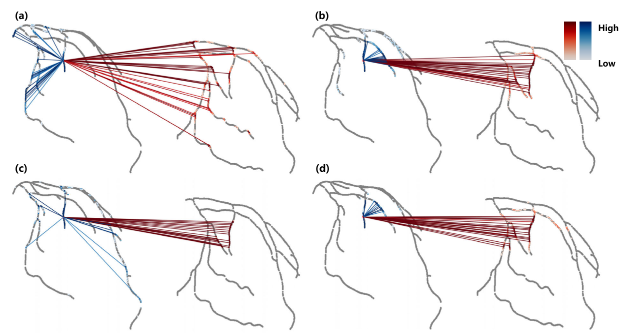

Fig. 8. Visualization of self (blue) and cross (red) attention scores. (a) and (b) represent shallower network layers focusing on broader node relationships, while (c) and (d) depict deeper layers, focusing on proximate and likely matching nodes.

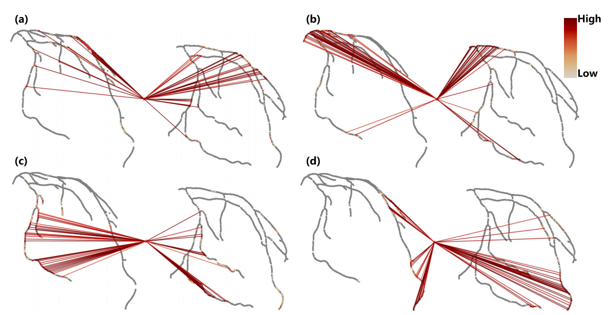

Fig. 9. Visualization of query attention scores. The QNG module not only clusters nodes based on their positions (b) but also clusters nodes based on bifurcation points (a) and vascular edges (c)(d).

the substantial performance enhancement provided by the deep supervision module. Augmenting centerline data with radius details boosts matching precision, while our novel data synthesis approach significantly enhances outcomes by effectively tackling non-rigid deformation issues in vascular matching. The proposed loss function has also improved network performance. Additionally, the incorporation of the QNG module, which clusters nodes and optimizes message passing, further improves the matching process, leading to increased precision and recall rates.

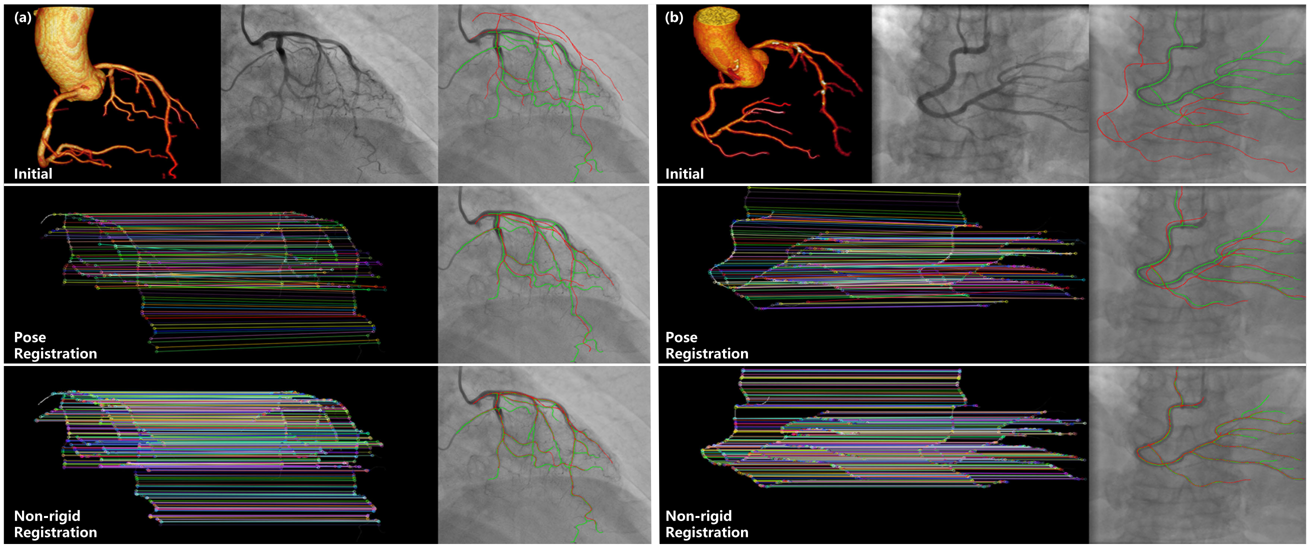

Fig. 10. The registration results of GVM-Net for left (a) and right (b) coronary arteries on clinical datasets. For each patient, the first row shows, from left to right, the CT coronary structure, the CAG image, and an overlaid image of the CAG centerline (green) with the projected CTA centerline (red). The second row shows the results of pose registration, and the third row presents the outcomes of non-rigid registration. The GVM-Net accurately and efficiently identifies numerous matched points for rigid and non-rigid transformations, with a mean matching time of 40 ms per case for 512 points. Zoom in on the figure to view the details.

TABLE VI
PERFORMANCE COMPARISON OF GVM-NET ON ORIGINAL AND INCOMPLETE SEGMENTATION DATASETS

Dataset/Metrics	mPD ↓	Precision ↑	Recall ↑	F1-Score ↑
Complete Segmentation	0.48	87.36	92.47	89.74
Incomplete Segmentation	0.53	86.56	92.14	89.16
P Value	0.29	0.18	0.42	0.20

TABLE VII
ABLATION EXPERIMENT OF QUERY NUMBER

Query Number/Metrics	mPD ↓	Precision ↑	Recall ↑	F1-Score ↑
No Query	0.52	86.56	90.37	88.32
Query 16	0.51	86.90	91.59	89.09
Query 25	**0.48**	**87.36**	**92.47**	**89.74**
Query 36	0.48	87.29	91.96	89.46
Query 64	0.47	87.39	91.92	89.50

1) Impact of Query Number: We performed ablation studies to analyze how the number of queries in the QNG module affects the results. Setting the initial queries to 16, 25, 36, and 64, with all other parameters unchanged, we developed four distinct models. The findings are detailed in Table VII. With 25 queries, GVM-Net achieves the best performance, achieving an mPD of 0.48 and an F1-score of 89.74%.

2) Importance of Loss Function: We conducted ablation tests to evaluate the effectiveness of the proposed loss function. The results are presented in Table VIII. Using focal loss improves precision to 85.56% and increases the F1-score to 87.47%. By integrating L_{Logdist} with a weight of 0.3, GVM-Net demonstrates best performance, achieving a precision of 86.56% and an F1-score of 88.32%.

3) Visualization of Graph Attention: GVM-Net draws inspiration from the graph attention network by representing

TABLE VIII
ABLATION EXPERIMENT OF LOSS FUNCTION. EVALUATES THE CROSS-ENTROPY (CE) LOSS, L_{FOCAL} AND L_{LOGDIST} VARIATIONS WITH $\beta = 0.1$, 0.3, 0.5 AND 1.0

Methods/Metrics	mPD ↓	Precision ↑	Recall ↑	F1-Score ↑
CE Loss	0.60	85.40	88.31	86.71
Focal Loss	0.61	85.56	89.70	87.47
Logdist-0.1	0.53	86.48	90.40	88.29
Logdist-0.3	**0.52**	**86.56**	**90.37**	**88.32**
Logdist-0.5	0.48	87.06	89.30	88.02
Logdist-1.0	0.50	86.98	88.28	87.39

vessel points as nodes in the graph and employing self- and cross-attention mechanisms for information propagation and updating. In this section, we visualize the attention scores between nodes as edge weights in the graph, depicted in Fig. 8. Furthermore, we employ attention scores between queries and nodes as clustering scores to visualize the clustering results of the QNG module, as shown in Fig. 9. The visualization results demonstrate that GVM-Net establishes a sparse, dynamic, and soft correspondence graph structure between nodes within and across images. Moreover, the query-based node clustering module not only aggregates and propagates information based on node positions, but also clusters vessels considering topology features such as bifurcation points and vascular edges, facilitating better matching of vascular points.

G. Validation on Clinical Data

We conducted quantitative validation on our manually annotated clinical dataset, comparing the performance of GMS, SuperGlue (finetuned), LightGlue (finetuned) and GVM-Net. Due to significant projection angle errors, we first used the results of the matching processes and the PnP algorithm for rigid registration. Subsequently, we performed

TABLE IX
QUANTITATIVE RESULTS ON TWO CLINICAL DATA GROUPS. STATISTICAL SIGNIFICANCE IS INDICATED BY ** (P < 0.01) AND * (P < 0.05)

Clinical Dataset	Method	F1-Score ↑	Precision ↑	Recall ↑	Error ↓
Group 1	GMS	52.65 ± 15.84**	72.85 ± 18.42**	42.93 ± 15.38**	1.79 ± 0.68**
	SuperGlue	77.98 ± 13.70**	82.29 ± 12.12	75.95 ± 17.33**	1.49 ± 0.29
	LightGlue	83.73 ± 7.18*	82.10 ± 9.93	87.14 ± 9.79	1.54 ± 0.29**
	GVM-Net	**85.79 ± 7.16**	**84.27 ± 9.18**	**87.99 ± 7.86**	**1.44 ± 0.28**
Group 2	GMS	49.85 ± 14.48**	71.07 ± 16.82*	40.30 ± 14.92**	1.88 ± 0.48**
	SuperGlue	71.95 ± 12.37**	77.66 ± 12.00	68.61 ± 15.02**	1.62 ± 0.34
	LightGlue	77.15 ± 10.35*	76.43 ± 11.47	79.28 ± 13.39	1.71 ± 0.29
	GVM-Net	**80.42 ± 8.05**	**79.32 ± 9.73**	**82.50 ± 10.04**	**1.63 ± 0.25**
Overall	GMS	51.38 ± 15.44**	72.04 ± 17.90**	41.73 ± 15.37**	1.83 ± 0.61**
	SuperGlue	75.24 ± 13.57**	80.19 ± 12.40	72.61 ± 16.88**	1.55 ± 0.32
	LightGlue	80.92 ± 9.72**	79.52 ± 11.12*	83.56 ± 12.32	1.62 ± 0.31*
	GVM-Net	**83.35 ± 8.11**	**82.02 ± 9.83**	**85.50 ± 9.41**	**1.52 ± 0.28**

feature matching and calculated the final metrics. All algorithms were configured with identical parameters: a maximum of 1024 keypoints and a match threshold of 0.1. The results are detailed in Table IX. The table indicates that GVM-Net shows a significant advantage in F1-score across both data groups when compared with other methods.

GVM-Net demonstrates superior accuracy in vascular matching tasks, particularly in challenging cases involving significant deformations and incomplete segmentations. Additionally, it effectively supports non-rigid 2D/3D coronary artery registration in clinical scenarios, as illustrated in Fig. 10. The workflow included extracting vessel centerlines from paired CTA and CAG images, projecting CTA centerlines onto the CAG plane, and optimizing camera pose using the PnP algorithm [56]. The projection error was set at 12 pixels, with a confidence level of 0.8 and a maximum of 5000 iterations. Subsequent non-rigid registration utilized the bSpline interpolation method from the Elastix library [57], with metrics including Euclidean Distance, Transform Bending Energy Penalty, and Transform Rigidity Penalty, weighted at a ratio of 1:0.7:0.7. The computational setup included an Intel Core i7-11700K CPU @ 3.60GHz, 32GB RAM, and an NVIDIA RTX A4000 GPU. Based on the results, GVM-Net can obtain 512 matches within 40 ms, facilitating rapid and accurate rigid or non-rigid registration of 2D/3D coronary arteries.

V. DISCUSSION AND CONCLUSION

In this study, we propose modeling the 2D/3D coronary artery registration problem as a centerline feature matching task and introduce GVM-Net, an end-to-end network for coronary vessel matching. GVM-Net comprises a graph attention network backbone and a correspondence prediction module. The former models topological relationships between nodes from different modalities using self-attention and cross-attention mechanisms efficiently. The QNG module further enhances this by grouping nodes in the feature space, resulting in more accurate correspondences and greater robustness in complex vascular scenarios. The latter resolves inconsistent vascular topological structures across modalities by adding redundant rows and columns in the matching matrix. The use of synthetic data for training GVM-Net effectively addresses the challenge of limited, high-quality clinical annotations, enabling robust supervised learning with precise ground truth. The data generation method is carefully designed to mimic real clinical complexities, including non-rigid vascular deformations, projection variability, topological inconsistencies, and noise, reducing the domain gap between synthetic and real-world data. Our proposed loss function also plays a crucial role in handling non-rigid deformations. Extensive experiments confirm the effectiveness of GVM-Net, even under challenging conditions such as false segmentation and large deformation, highlighting its promise for clinical applications.

The rich topological information of vascular trees presents both opportunities and challenges for registration. However, previous methods in the field suffer from a lack of effective utilization of vascular topological information, leading to slow registration speeds and difficulties in resolving topological inconsistencies. In recent years, several methods have made notable progress by leveraging vascular tree topology for registration, including those proposed by Wu et al. [20] and Zhu et al. [21]. These methods can achieve high accuracy when vessel segments are correctly matched. However, they rely heavily on the construction and matching of vascular graph structures, which introduces several challenges. When there are inconsistencies in vascular topology across different modalities or discontinuities in the vessel segmentation results, these methods often struggle to accurately match the vessel segments [21], leading to reduced registration accuracy. Additionally, constructing the vascular topology is a complex task, particularly in 2D centerline images where vessel crossings and overlaps often result in false bifurcations or loops. To the best of our knowledge, no method can fully reconstruct the correct vascular topology from 2D centerlines, which further complicates graph-based matching. Moreover, in cases with numerous vessels and bifurcations, graph matching can become time-consuming, making registration in complex vascular structures even more challenging [20], [21]. In contrast, we propose abandoning the complex process of constructing and matching vessel graph in traditional methods. By combining graph attention networks, GVM-Net achieves dense correspondences between vessels, yielding significant improvements in both efficiency and robustness. We visualize the self-attention and cross-attention scores between nodes and find that GVM-Net can establish rich and interpretable topological relationships between nodes within and across modalities, leading to accurate matching results. Our experiments also demonstrate that GVM-Net maintains high accuracy even in the presence of segmentation discontinuities, giving it a clear advantage in handling real-world clinical data where vascular segmentation is often imperfect. Furthermore, we believe that GVM-Net could also complement the methods proposed by Wu et al. and Zhu et al. by providing initial matching points, which could greatly simplify the process of vessel segment matching, thereby improving the overall performance of these approaches.

Another widely used method in this field is point cloud-based registration. These methods iteratively establish point correspondences and use them to compute non-rigid transformation parameters for registration. However, point clouds typically contain only positional information and lack local context. To address this, Belongie et al. [29] proposed the use of shape context descriptors, which incorporate the neighborhood structure of the point set to improve the accuracy of point correspondences. However, these methods tend to perform poorly when the point cloud contains significant noise or outliers. Additionally, traditional iterative point cloud registration methods are computationally intensive. In our experiments, we compared GVM-Net with several point cloud-based methods, including Mixture TPS-RPM, TPS-L2, TPS-KC, RPM-L2E, and PR-GLS. The results demonstrated that GVM-Net significantly outperforms these methods in both computational efficiency and registration accuracy. Notably, GVM-Net achieved superior performance in challenging scenarios involving outliers, noise, and complex vessel structures, showcasing its robustness and efficiency compared to traditional point cloud registration approaches.

In recent years, deep learning-based nonrigid point cloud registration methods have rapidly advanced, such as Lepard [58] and GraphSCNet [59]. These methods use powerful backbone networks to encode local geometric features of point clouds, and further employ attention mechanisms to achieve point matching and registration. While these methods have demonstrated strong performance in general point cloud registration tasks, their efficacy in the vascular registration domain remains largely unexplored. Moreover, converting vessel centerlines into point clouds and then re-extracting local features using backbone networks inherently leads to a loss of texture information compared to directly extracting keypoints and descriptors from the centerline images. We will explore these methods in future research to assess their applicability and performance in the vascular registration domain.

Despite the numerous advantages of GVM-Net, our proposed method still has some limitations. First, the clinical dataset used for testing is relatively small, and only sparse manually labeled correspondences are available for each data pair. Nevertheless, GVM-Net demonstrated significant advantages in the F1-score compared with other methods. Although SuperGlue achieved a comparable matching error to GVM-Net, this was mainly due to its relatively lower recall rate, as the error was calculated only for recalled matching points. In the future, we aim to expand our clinical dataset and improve labeling efficiency and accuracy with improved tools. Second, the vascular structure needs to be segmented from the image. Although we have validated GVM-Net's robustness to incomplete segmentation results, the complexity of vascular structures still poses high demands on the segmentation algorithms. Third, exploring direct matching of 2D and 3D points without the need for a projection process is also one of the future research directions.

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References

[1] C. Cooper, "Global, regional, and national disability-adjusted life-years (DALY) for 359 diseases and injuries and health life expectancy (HALE) for 195 countries and territories, 1990–2017: A systematic analysis for the Global Burden of Disease Study 2017," *Lancet*, vol. 392, pp. 1859–1922, Nov. 2018.

[2] B. B. Ghoshhajra et al., "Real-time fusion of coronary CT angiography with X-ray fluoroscopy during chronic total occlusion PCI," *Eur. Radiol.*, vol. 27, no. 6, pp. 2464–2473, Jun. 2017.

[3] A. Rolf et al., "Preprocedural coronary CT angiography significantly improves success rates of PCI for chronic total occlusion," *Int. J. Cardiovascular Imag.*, vol. 29, no. 8, pp. 1819–1827, Dec. 2013.

[4] G. S. Werner, "Use of coronary computed tomographic angiography to facilitate percutaneous coronary intervention of chronic total occlusions," *Circulat., Cardiovascular Intervent.*, vol. 12, Oct. 2019, Art. no. e007387.

[5] C. Collet et al., "Implementing coronary computed tomography angiography in the catheterization laboratory," *JACC, Cardiovascular Imag.*, vol. 14, no. 9, pp. 1846–1855, Sep. 2021.

[6] C. Gatta, S. Balocco, V. Martin-Yuste, R. Leta, and P. Radeva, "Non-rigid multi-modal registration of coronary arteries using SIFTflow," in *Pattern Recognition and Image Analysis*. Germany: Springer, Jan. 2011, pp. 159–166.

[7] A. R. Gouveia, C. Metz, L. Freire, P. Almeida, and S. Klein, "Registration-by-regression of coronary CTA and X-ray angiography," *Comput. Methods Biomech. Biomed. Eng., Imag. Visualizat.*, vol. 5, no. 3, pp. 208–220, May 2017.

[8] E. Serradell, A. Romero, R. Leta, C. Gatta, and F. Moreno-Noguer, "Simultaneous correspondence and non-rigid 3D reconstruction of the coronary tree from single X-ray images," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 850–857.

[9] M. Groher, D. Zikic, and N. Navab, "Deformable 2D-3D registration of vascular structures in a one view scenario," *IEEE Trans. Med. Imag.*, vol. 28, no. 6, pp. 847–860, Jun. 2009.

[10] N. Baka, C. T. Metz, C. J. Schultz, R.-J. van Geuns, W. J. Niessen, and T. van Walsum, "Oriented Gaussian mixture models for nonrigid 2D/3D coronary artery registration," *IEEE Trans. Med. Imag.*, vol. 33, no. 5, pp. 1023–1034, May 2014.

[11] T. Benseghir, G. Malandain, and R. Vaillant, "Iterative closest curve: A framework for curvilinear structure registration application to 2D/3D coronary arteries registration," in *Advanced Information Systems Engineering*. Berlin, Germany: Springer, Jan. 2013, pp. 179–186.

[12] T. Benseghir, G. Malandain, and R. Vaillant, "A tree-topology preserving pairing for 3D/2D registration," *Int. J. Comput. Assist. Radiol. Surg.*, vol. 10, no. 6, pp. 913–923, Jun. 2015.

[13] E. Serradell, M. A. Pinheiro, R. Sznitman, J. Kybic, F. Moreno-Noguer, and P. Fua, "Non-rigid graph registration using active testing search," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 3, pp. 625–638, Mar. 2015.

[14] E. Serradell, P. Glowacki, J. Kybic, F. Moreno-Noguer, and P. Fua, "Robust non-rigid registration of 2D and 3D graphs," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 996–1003.

[15] M. A. Pinheiro, J. Kybic, and P. Fua, "Geometric graph matching using Monte Carlo tree search," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 11, pp. 2171–2185, Nov. 2017.

[16] S. Liu, P. Liu, Z. Li, Y. Zhang, W. Li, and X. Tang, "A 3D/2D registration of the coronary arteries based on tree topology consistency matching," *Biomed. Signal Process. Control*, vol. 38, pp. 191–199, Sep. 2017.

[17] J. Zhu et al., "Heuristic tree searching for pose-independent 3D/2D rigid registration of vessel structures," *Phys. Med. Biol.*, vol. 65, no. 5, Mar. 2020, Art. no. 055010.

[18] J. Zhu et al., "Iterative closest graph matching for non-rigid 3D/2D coronary arteries registration," *Comput. Methods Programs Biomed.*, vol. 199, Feb. 2021, Art. no. 105901.

[19] H. Zhang, J. Zhang, W. Wu, H. Xie, and L. Gu, "Deformable registration of coronary arteries with topological constraints for image-guided vascular interventions," in *Proc. 42nd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2020, pp. 1347–1350.

[20] W. Wu, J. Zhang, W. Peng, H. Xie, S. Zhang, and L. Gu, "CAR-net: A deep learning-based deformation model for 3D/2D coronary artery registration," *IEEE Trans. Med. Imag.*, vol. 41, no. 10, pp. 2715–2727, Oct. 2022.

[21] J. Zhu et al., "3D/2D vessel registration based on Monte Carlo tree search and manifold regularization," *IEEE Trans. Med. Imag.*, vol. 43, no. 5, pp. 1727–1739, May 2024.
[22] H. Chui and A. Rangarajan, "A new algorithm for non-rigid point matching," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, vol. 2, Jun. 2000, pp. 44–51.
[23] Y. Zheng and D. Doermann, "Robust point matching for nonrigid shapes by preserving local neighborhood structures," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 28, no. 4, pp. 643–649, Apr. 2006.
[24] J. Ma, J. Zhao, and A. L. Yuille, "Non-rigid point set registration by preserving global and local structures," *IEEE Trans. Image Process.*, vol. 25, no. 1, pp. 53–64, Jan. 2016.
[25] J. Ma, J. Zhao, J. Tian, Z. Tu, and A. L. Yuille, "Robust estimation of nonrigid transformation for point set registration," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 2147–2154.
[26] X. Zhang et al., "SPR-net: Structural points based registration for coronary arteries across systolic and diastolic phases," in *Medical Image Computing and Computer Assisted Intervention—MICCAI*, H. Greenspan, A. Madabhushi, P. Mousavi, S. Salcudean, J. Duncan, T. Syeda-Mahmood, and R. Taylor, Eds., Cham, Switzerland: Springer, 2023, pp. 791–801.
[27] B. Jian and B. C. Vemuri, "A robust algorithm for point set registration using mixture of Gaussians," in *Proc. 10th IEEE Int. Conf. Comput. Vis. (ICCV)*, vol. 2, Oct. 2005, pp. 1246–1251.
[28] Y. Tsin and T. Kanade, "A correlation-based approach to robust point set registration," in *Computer Vision—ECCV*, T. Pajdla and J. Matas, Eds., Berlin, Germany: Springer, 2004, pp. 558–569.
[29] S. Belongie, J. Malik, and J. Puzicha, "Shape matching and object recognition using shape contexts," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 4, pp. 509–522, Apr. 2002.
[30] S. Matl, R. Brosig, M. Baust, N. Navab, and S. Demirci, "Vascular image registration techniques: A living review," *Med. Image Anal.*, vol. 35, pp. 1–17, Jan. 2017.
[31] A. Sotiras, C. Davatzikos, and N. Paragios, "Deformable medical image registration: A survey," *IEEE Trans. Med. Imag.*, vol. 32, no. 7, pp. 1153–1190, Jul. 2013.
[32] P. Markelj, D. Tomaževič, B. Likar, and F. Pernuš, "A review of 3D/2D registration methods for image-guided interventions," *Med. Image Anal.*, vol. 16, no. 3, pp. 642–661, Apr. 2012.
[33] A. Q. Wang, E. M. Yu, A. V. Dalca, and M. R. Sabuncu, "A robust and interpretable deep learning framework for multi-modal registration via keypoints," *Med. Image Anal.*, vol. 90, Dec. 2023, Art. no. 102962.
[34] N. Li, J. He, Y. Chen, and S. Zhou, "A survey on the progress of computer-assisted vascular intervention," *J. Comput.-Aided Des. Comput. Graph.*, vol. 34, no. 7, pp. 985–1010, Jul. 2022.
[35] R. A. McLaughlin, J. Hipwell, D. J. Hawkes, J. A. Noble, J. V. Byrne, and T. C. Cox, "A comparison of a similarity-based and a feature-based 2-D-3-D registration method for neurointerventional use," *IEEE Trans. Med. Imag.*, vol. 24, no. 8, pp. 1058–1066, Aug. 2005.
[36] S. Habert, P. Khurd, and C. Chefd'hotel, "Registration of multiple temporally related point sets using a novel variant of the coherent point drift algorithm: Application to coronary tree matching," *Proc. SPIE*, vol. 8669, pp. 157–167, Mar. 2013.
[37] A. Rangarajan et al., "A robust point-matching algorithm for autoradiograph alignment," *Med. Image Anal.*, vol. 1, no. 4, pp. 379–398, Sep. 1997.
[38] S. Gold, A. Rangarajan, C.-P. Lu, S. Pappu, and E. Mjolsness, "New algorithms for 2D and 3D point matching: Pose estimation and correspondence," *Pattern Recognit.*, vol. 31, no. 8, pp. 1019–1031, 1998.
[39] B. Jian and B. C. Vemuri, "Robust point set registration using Gaussian mixture models," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 8, pp. 1633–1645, Aug. 2011.
[40] S. Yoon, C. H. Yoon, and D. Lee, "Topological recovery for non-rigid 2D/3D registration of coronary artery models," *Comput. Methods Programs Biomed.*, vol. 200, Mar. 2021, Art. no. 105922.
[41] Y. Wang, Y. Sun, Z. Liu, S. E. Sarma, M. M. Bronstein, and J. M. Solomon, "Dynamic graph CNN for learning on point clouds," *ACM Trans. Graph.*, vol. 38, no. 5, pp. 1–12, 2019.
[42] J. Ma, X. Jiang, A. Fan, J. Jiang, and J. Yan, "Image matching from handcrafted to deep features: A survey," *Int. J. Comput. Vis.*, vol. 129, no. 1, pp. 23–79, Jan. 2021.
[43] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperGlue: Learning feature matching with graph neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4937–4946.
[44] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30. Red Hook, NY, USA: Curran Associates, Jun. 2017, pp. 5998–6008.
[45] G. Peyré and M. Cuturi, "Computational optimal transport: With applications to data science," *Found. Trends Mach. Learn.*, vol. 11, nos. 5–6, pp. 355–607, 2019.
[46] P. Lindenberger, P. E. Sarlin, and M. Pollefeys, "LightGlue: Local feature matching at light speed," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 17627–17638.
[47] D. Jha et al., "ResUNet++: An advanced architecture for medical image segmentation," in *Proc. IEEE Int. Symp. Multimedia (ISM)*, Dec. 2019, pp. 225–2255.
[48] Ö. Çiçek, A. Abdulkadir, S. S. Lienkamp, T. Brox, and O. Ronneberger, "3D U-Net: Learning dense volumetric segmentation from sparse annotation," in *Medical Image Computing and Computer-Assisted Intervention—MICCAI*, S. Ourselin, L. Joskowicz, M. R. Sabuncu, G. Unal, and W. Wells, Eds., Cham, Switzerland: Springer, 2016, pp. 424–432.
[49] R. Azad et al., "Medical image segmentation review: The success of U-Net," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 10076–10095, Dec. 2024.
[50] T. Y. Zhang and C. Y. Suen, "A fast parallel algorithm for thinning digital patterns," *Commun. ACM*, vol. 27, no. 3, pp. 236–239, Mar. 1984.
[51] T. C. Lee, R. L. Kashyap, and C. N. Chu, "Building skeleton models via 3-D medial surface axis thinning algorithms," *CVGIP, Graph. Models Image Process.*, vol. 56, no. 6, pp. 462–478, Nov. 1994.
[52] D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperPoint: Self-supervised interest point detection and description," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 224–236.
[53] J. Sun, Z. Shen, Y. Wang, H. Bao, and X. Zhou, "LoFTR: Detector-free local feature matching with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8918–8927.
[54] M. Muja and D. G. Lowe, "Fast approximate nearest neighbors with automatic algorithm configuration," in *Proc. Int. Conf. Comput. Vis. Theory Appl.*, vol. 2, Feb. 2009, pp. 331–340.
[55] J.-W. Bian et al., "GMS: Grid-based motion statistics for fast, ultra-robust feature correspondence," *Int. J. Comput. Vis.*, vol. 128, no. 6, pp. 1580–1593, Jun. 2020.
[56] G. Terzakis and M. Lourakis, "A consistently fast and globally optimal solution to the perspective-n-point problem," in *Proc. 16th Eur. Conf.*, Glasgow, U.K. Berlin, Germany: Springer, Aug. 2020, pp. 478–494.
[57] S. Klein, M. Staring, K. Murphy, M. A. Viergever, and J. P. W. Pluim, "Elastix: A toolbox for intensity-based medical image registration," *IEEE Trans. Med. Imag.*, vol. 29, no. 1, pp. 196–205, Jan. 2010.
[58] Y. Li and T. Harada, "Lepard: Learning partial point cloud matching in rigid and deformable scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5544–5554.
[59] Z. Qin, H. Yu, C. Wang, Y. Peng, and K. Xu, "Deep graph-based spatial consistency for robust non-rigid point cloud registration," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 5394–5403.